

CS1006

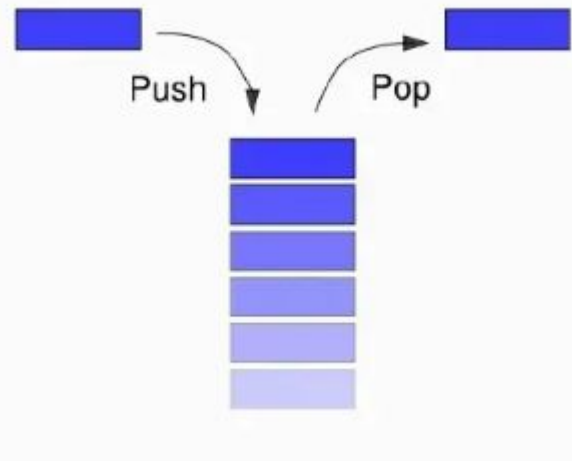
Data Structures

Unit – III

Stacks and Queues

Introduction – Stacks

- A Stack is a list in which insertion and deletion take place at the same end known as top.
- Stack follows **LIFO** strategy -Last In First Out
- The last element inserted will be the first to be retrieved.



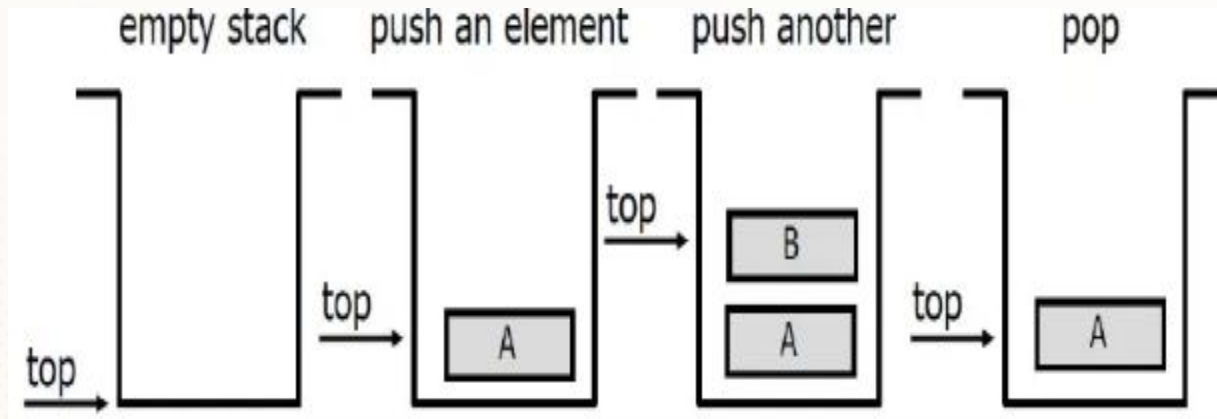
Stack Operations

1. Creating a Stack
2. Checking Stack as either Empty or Full
3. Inserting element into Stack – Push()
4. Deleting element from Stack – Pop()
5. Accessing the Top element of Stack
6. Display the elements of Stack

Stack Operations

Primary Operations – Push and Pop

- ❖ **Push** - Adding an element to the Top of Stack
- ❖ **Pop** – Removing an element from the Top of Stack



- ❖ **Top:** A Pointer that points the top element in the Stack.
- ❖ **Stack Underflow :** When there is no element in the Stack, the status is called Stack Underflow.
- ❖ **Stack Overflow:** When the Stack contains equal number of elements as per its capacity and no more elements can be added, then the status is called Stack Overflow.

Stack Implementation

Implementation can be done in two ways-

❖ Static Implementation

- ✓ Stack has fixed size and implemented using arrays
- ✓ Inefficient for memory utilization

❖ Dynamic Implementation

- ✓ Stack grow in size as needed and implemented as linked list.
- ✓ Done through pointers
- ✓ Efficient utilization of memory during insertion and deletion

Applications of Stack

- Pile of Books, Plate Trays, Evaluating Expressions, Program Execution Stack, Reversing Strings
- Back and Forward Buttons in Web browser
- Power point Application's Undo and Redo Commands
- Java uses Stacks to execute operations in program
- Many other data structure also rely on stacks



Arithmetic Expressions

Prefix Notation-It is a form of notation which places its operators to the left of their operands. It is also known as Polish Notation.

Syntax: [operator] [operand1] [operand2]

Example: $- * + A B C / D + E F$

Infix Notation- It is a form of notation which places its operator between its operands.

Syntax: [operand1] [operator] [operand2]

Example: $(A + B) * C - D / (E + F)$

Postfix Notation-It is a form of notation which places its operator comes after its operands. It is also known as Reverse Polish Notation (RPN) or Suffix Notation. It enables easy evaluation of expressions.

Syntax: [operand1] [operand2] [operator]

Example: $A B + C * D E F + / -$

Evaluating Arithmetic Expressions

Infix into postfix expression

Infix	Postfix
$2 + 3$	$2\ 3\ +$
$2 + 3 * 6$	$3\ 6\ * \ 2\ +$
$(2 + 3) * 6$	$2\ 3\ + \ 6\ *$
$A / (B * C) + D * E - A * C$	$A\ B\ C\ * \ / \ D\ E\ * \ + \ A\ C\ * \ -$

Infix to Postfix Expression

Let Q be the arithmetic expression written in Infix notation. The algorithm finds the equivalent postfix expression P.

Steps Followed:

1. Push "(" onto stack and add ")" to the end of Q.
2. Scan Q from left to right and repeat steps 3 to 6 for each element of Q until the stack is empty.
3. If an operand is encountered, add it to P.
4. If a left parenthesis is encountered, push it onto stack.
5. If an operator is encountered, then
 - a. Repeatedly pop from stack and add to P each operator (on the top of stack) which has same precedence or as higher precedence than operator.
 - b. Add operator to stack.
6. If a right parenthesis is encountered, then
 - a. Repeatedly pop from stack and add to P each operator (on the top of stack) until a left parenthesis is encountered.
 - b. Remove the left parenthesis
7. Exit

Infix to Postfix Expression Conversion

Convert $((A - (B + C)) * D) \uparrow (E + F)$ infix expression to postfix form:

SYMBOL	POSTFIX STRING	STACK	REMARKS
(		(	
(		((	
A	A	((	
-	A	((-	
(	A	((- (	
B	A B	((- (	
+	A B	((- (+	
C	A B C	((- (+	
)	A B C +	((-	
)	A B C + -	(	
*	A B C + -	(*	
D	A B C + - D	(*	
)	A B C + - D *		
↑	A B C + - D *	↑	
(	A B C + - D *	↑ (	
E	A B C + - D * E	↑ (	
+	A B C + - D * E	↑ (+	
F	A B C + - D * E F	↑ (+	
)	A B C + - D * E F +	↑	
End of string	A B C + - D * E F + ↑	The input is now empty. Pop the output symbols from the stack until it is empty.	

Evaluation of Postfix Expression

Let P is an arithmetic expression written in postfix notation. The algorithm uses stack to hold operands, evaluates P.

Steps Followed :

1. Add a right parenthesis “)” at the end of P
2. Scan P from left to right and repeat steps 3 and 4 for each element of P until the sentinel “)” is encountered.
3. If an operand is encountered, put it on stack.
4. If an operator is encountered then,
 - a. Remove the two top element of stack
 - b. Evaluate the two operands with operator
 - c. Place the result back on to stack
5. Set value equal to top element of stack
6. Exit

Evaluation of Postfix Expression

Postfix notation P : 5, 6, 2, +, *, 12, 4, /, -)

Symbol Scanned	STACK
(1) 5	5
(2) 6	5, 6
(3) 2	5, 6, 2
(4) +	5, 8
(5) *	40
(6) 12	40, 12
(7) 4	40, 12, 4
(8) /	40, 3
(9) -	37
(10))	

Introduction to Queues

- Queue is a Linear Data Structure.
- It follows **FIFO**-First In First Out Principle.
- It has two pointers-Front and Rear.
- Elements are inserted at one end called Front end and deleted at another end called Rear end.



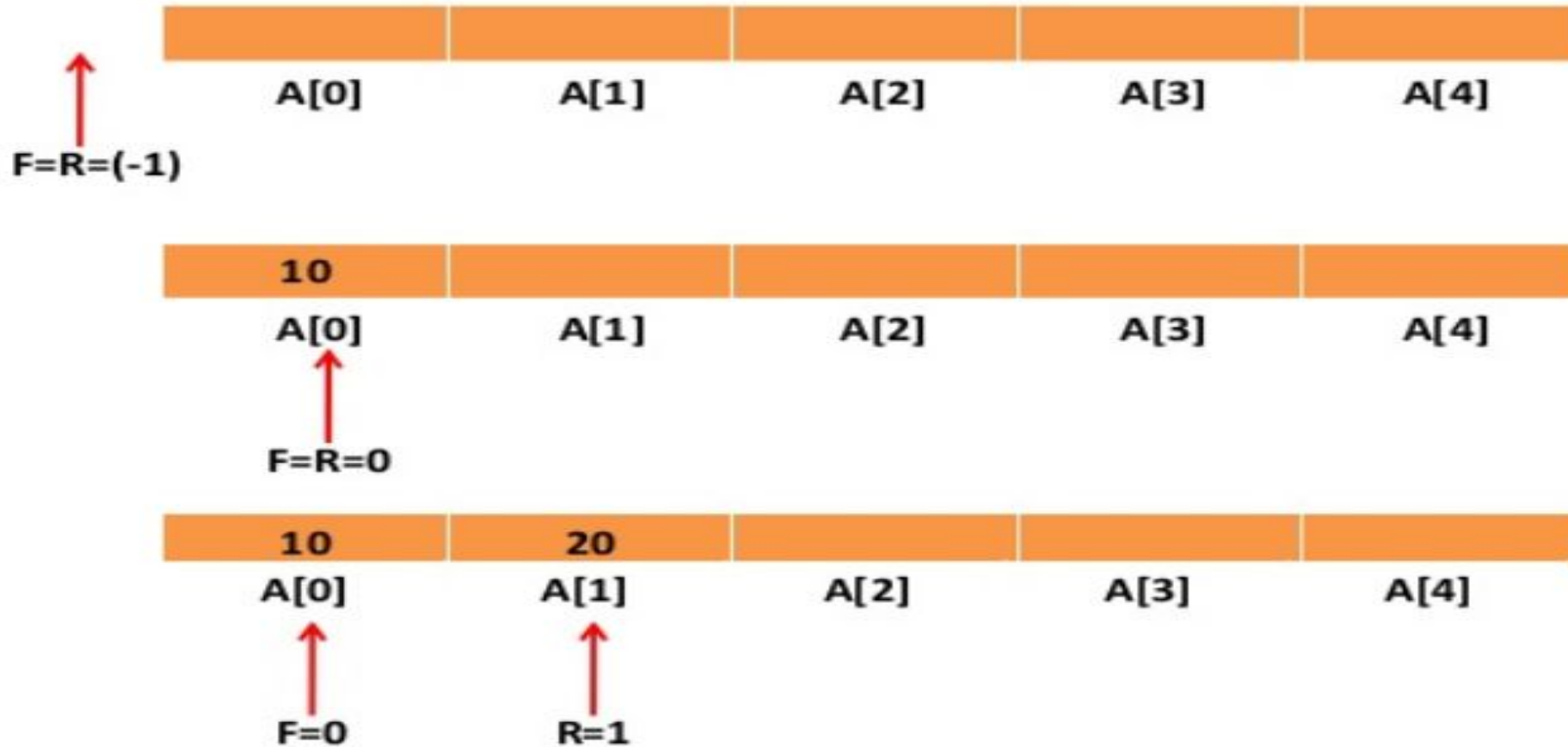
Primary Operations

- **Enqueue**- Inserting an element into the queue at rear end
- **Dequeue** – Deleting an element from the Queue at front end.

Insertion Algorithm

```
Step 1: If REAR = MAX – 1 then  
        Write “Queue is Overflow”  
        Goto step 4  
        [End of IF]  
Step 2: IF FRONT=-1 and REAR=-1  
        SET FRONT=REAR=0  
        ELSE  
        SET REAR=REAR+1  
        [END OF IF]  
Step 3: SET QUEUE [REAR] = NUM  
Step 4: EXIT
```

Insertion - Example

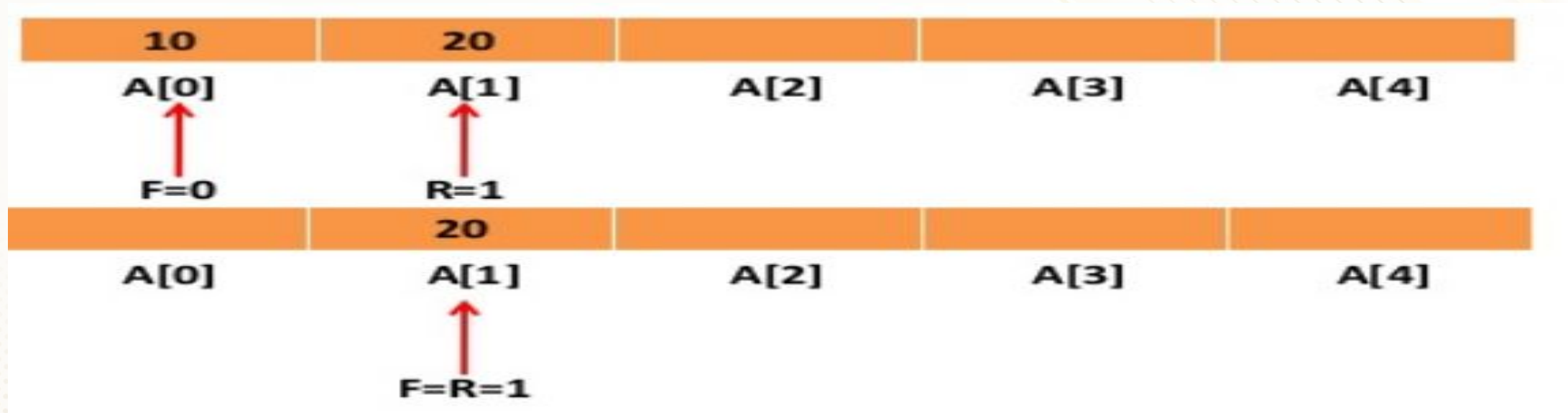


Deletion Algorithm

```
Step 1: IF FRONT = -1 OR FRONT > REAR  
        Write "Queue is Underflow"  
    ELSE  
        SET VAL = QUEUE [FRONT]  
        FRONT = FRONT + 1  
    [END OF IF]
```

```
Step 2: EXIT
```

Deletion - Example



Types of Queues

Some of the Queue's are:

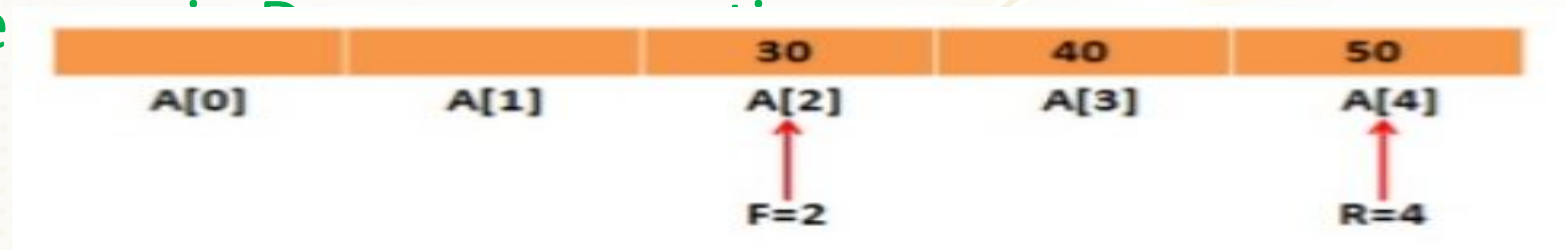
1. Circular Queue
2. Priority Queue
3. Deque
4. Multiple Queue

Circular Queue

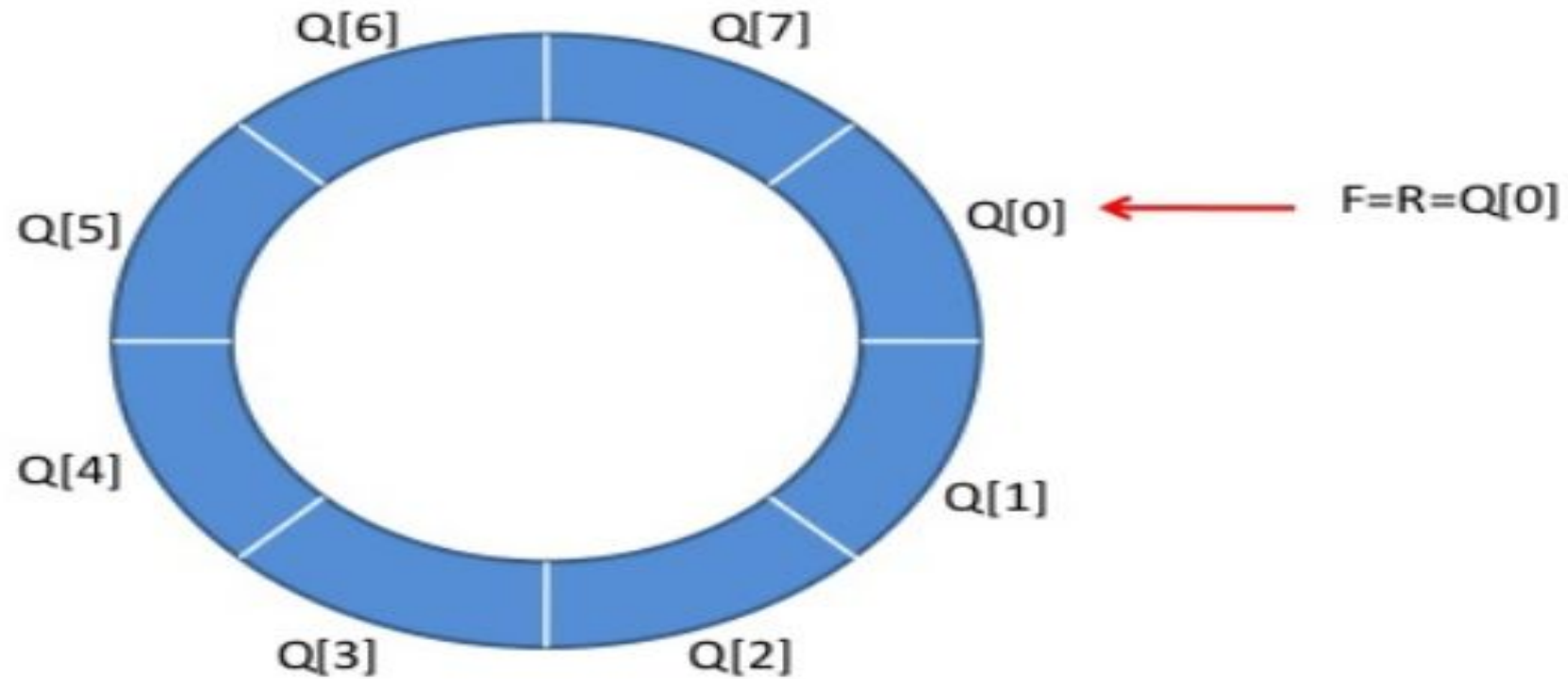
- The arrangement of elements $Q[0], Q[1], \dots, Q[n]$ in a circular fashion with $Q[1]$ following $Q[n]$ is called Circular Queue.
- The last node is connected to first node to make a circle.
- Initially, both Front and Rear Pointers point to the beginning of the array.
- It is also known as “Ring Buffer”.

Need for Circular Queue:

❖ Wastage of memory



Circular Queue – Example



Circular Queue – Enqueue

Step 1: IF FRONT=0 and REAR=MAX-1 Then
Write "Overflow"
Goto Step 4

[END OF IF]

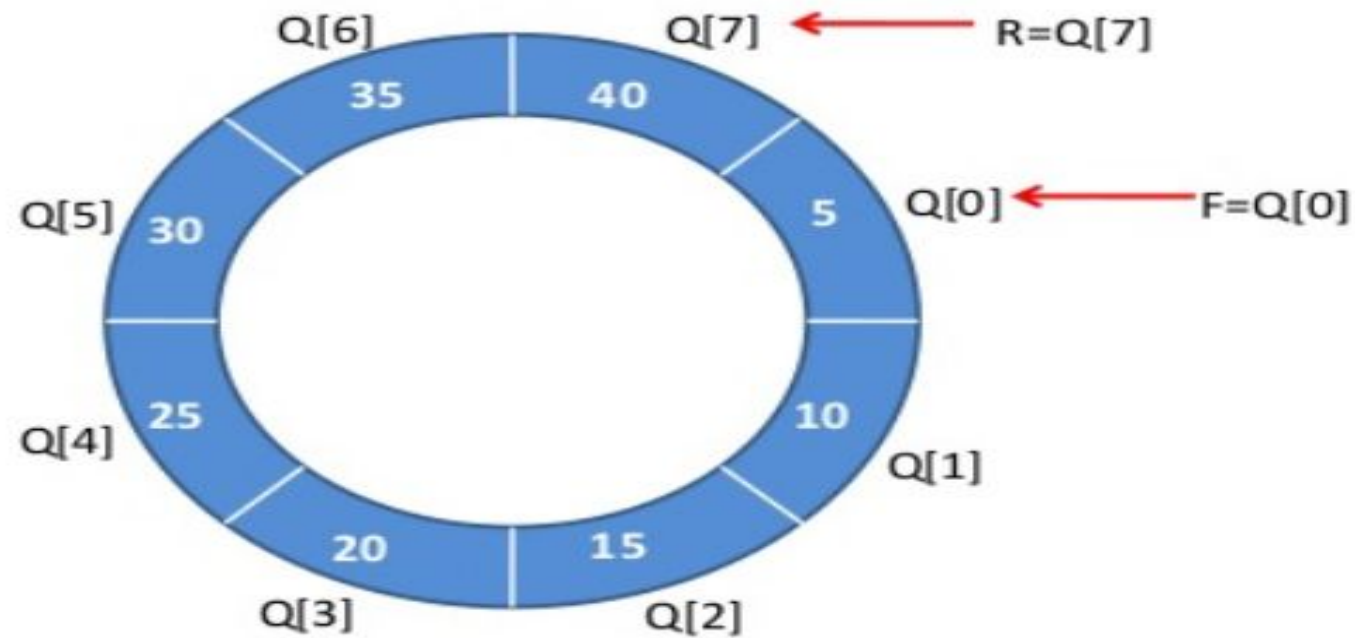
Step 2: IF FRONT = REAR=-1 then
SET FRONT=REAR=0
ELSE IF REAR=MAX-1 and FRONT!=0
SET REAR=0
ELSE
SET REAR=REAR+1

[END OF IF]

Step 3: SET QUEUE[REAR]=VAL

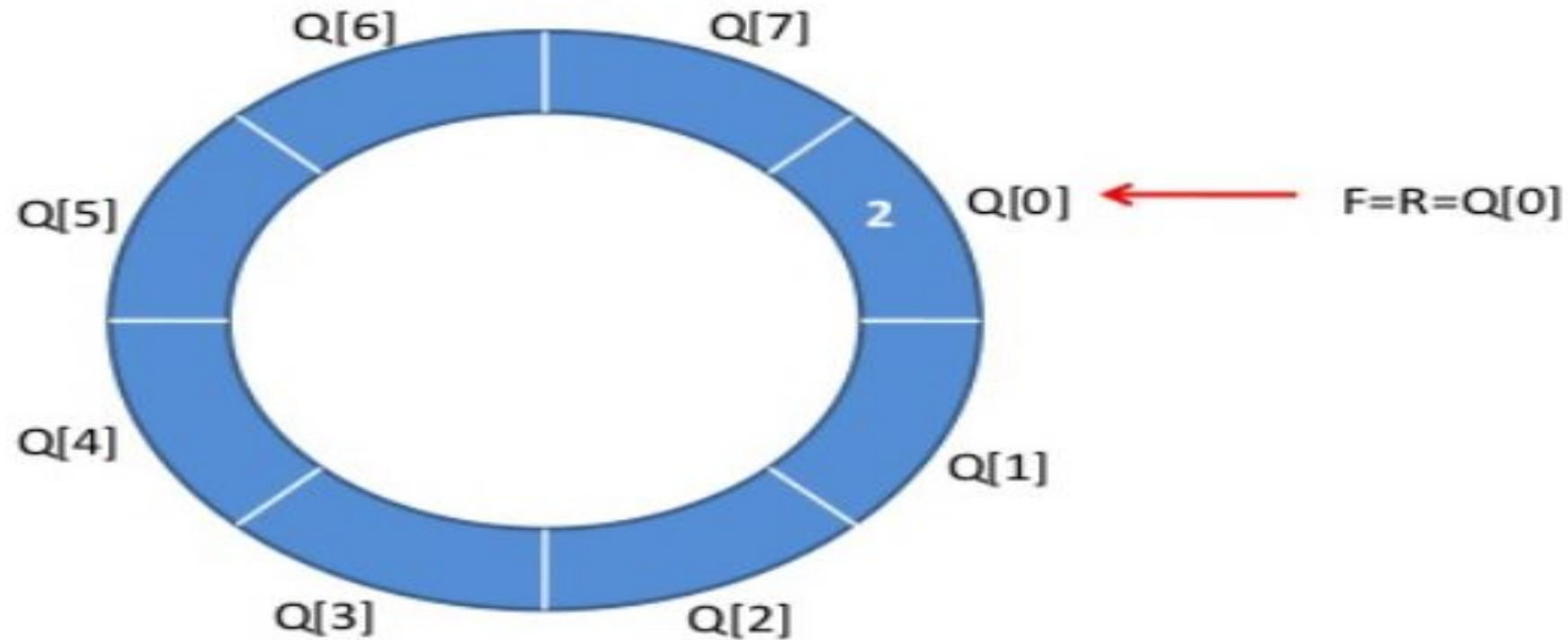
Step 4: EXIT

Enqueue Operation



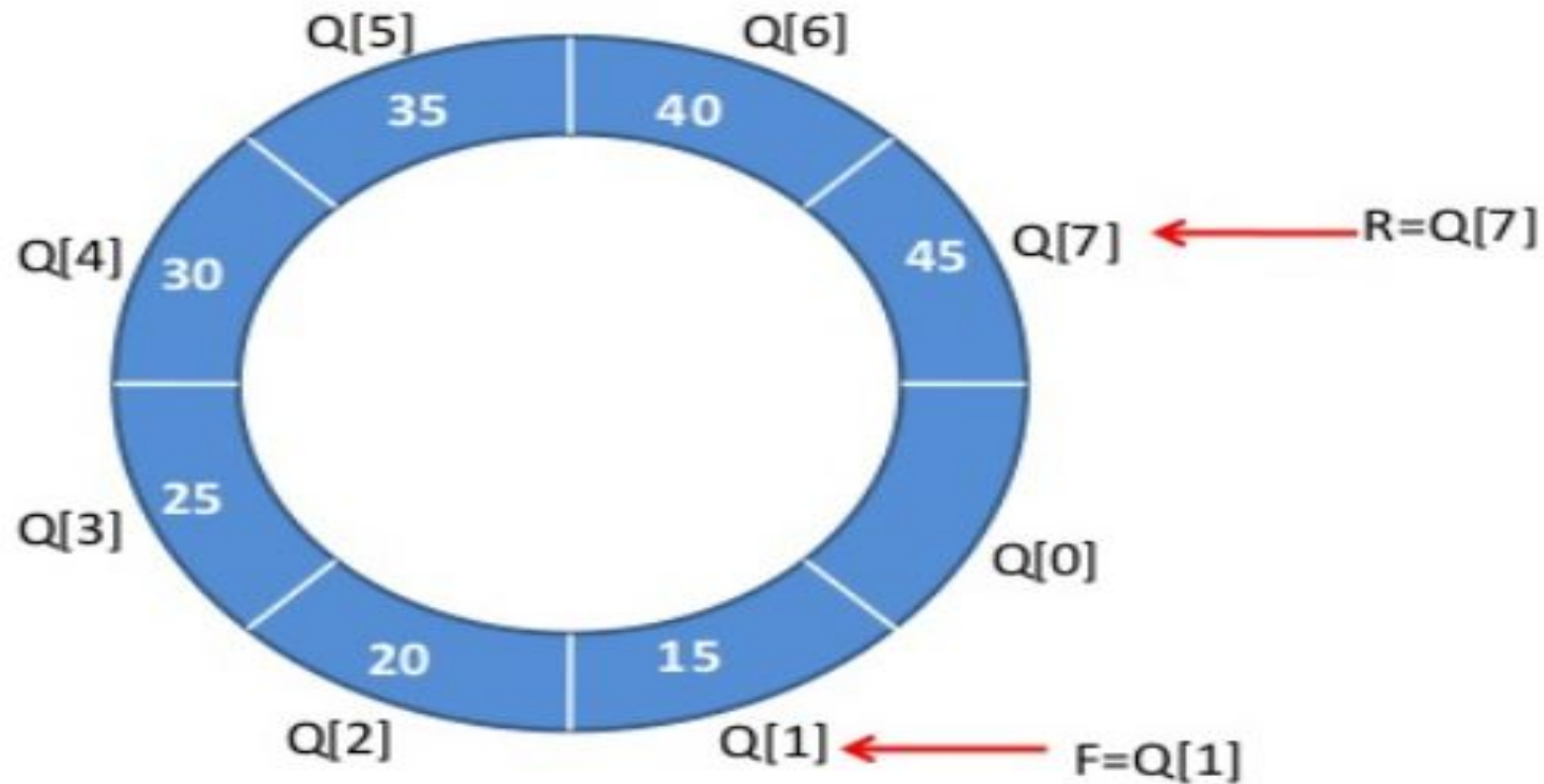
Queue is full(Overflow)

Enqueue Operation

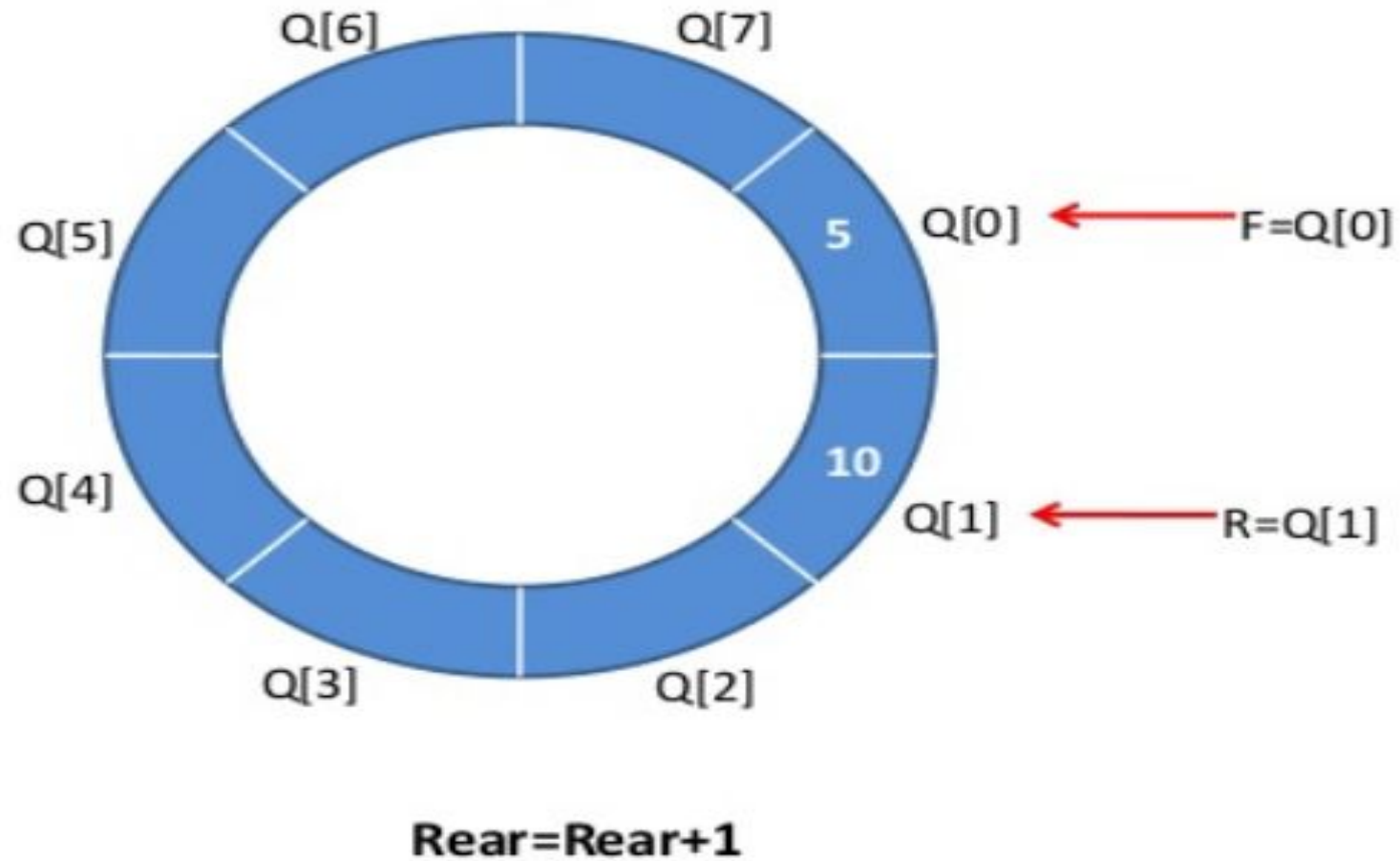


If $F=R=-1$ then $F=R=0$

Enqueue Operation



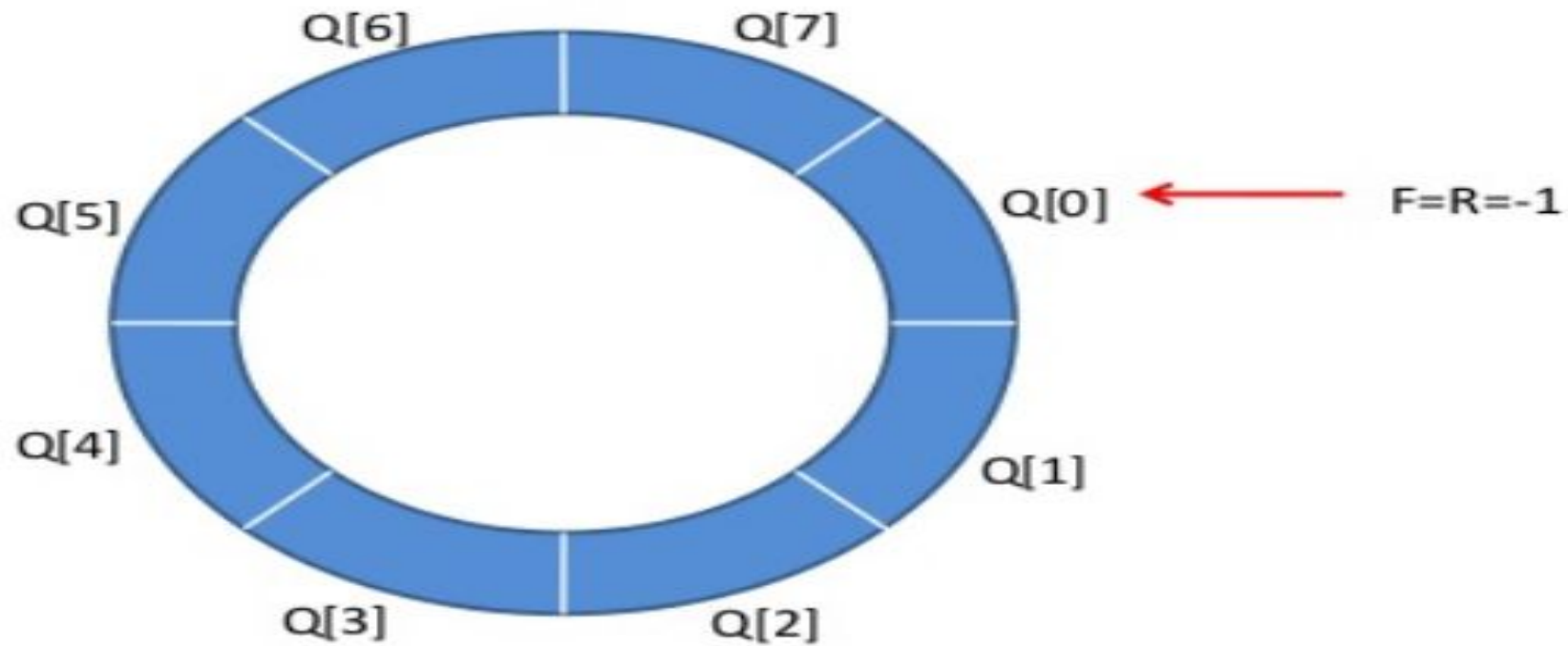
Enqueue Operation



Circular Queue – Dequeue

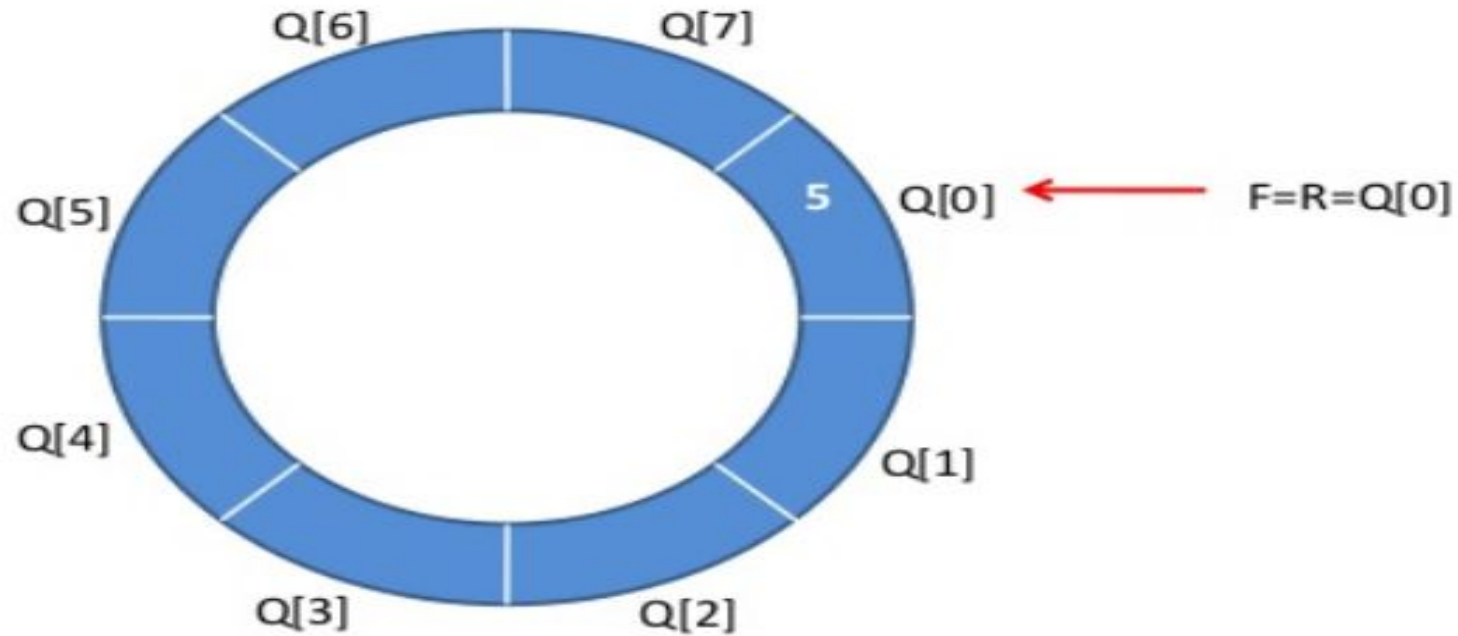
```
Step 1: If FRONT = -1 then
        Write ("Circular Queue Underflow")
        GOTO Step 4
Step 2: SET VAL=QUEUE[FRONT]
Step 3: If FRONT = REAR then
        SET FRONT=REAR=-1
    ELSE
        IF FRONT=MAX-1
            SET FRONT=0
        ELSE
            SET FRONT=FRONT+1
        [END OF IF]
    [END OF IF]
Step 4: EXIT
```

Deque Operation



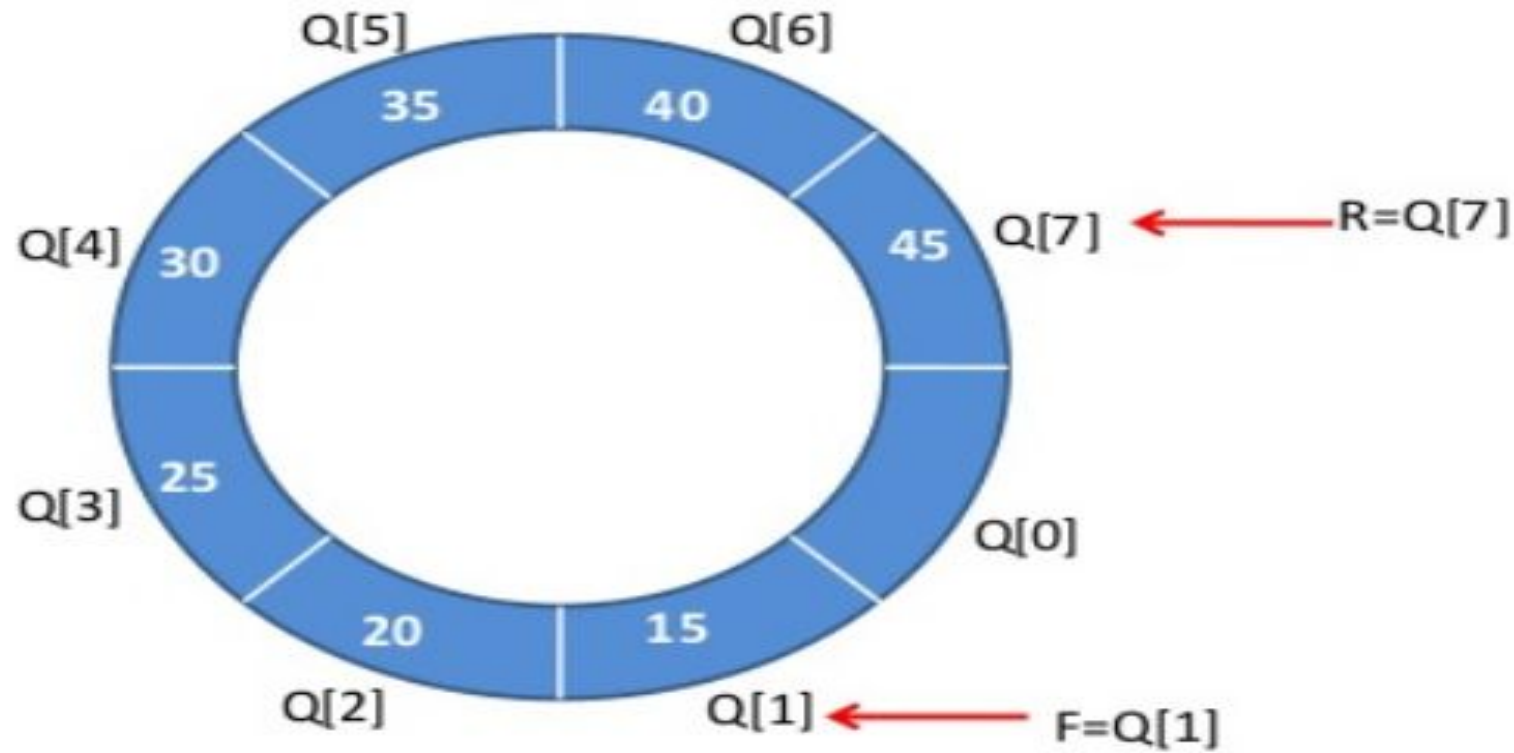
Queue is Empty(Underflow)

Deque Operation



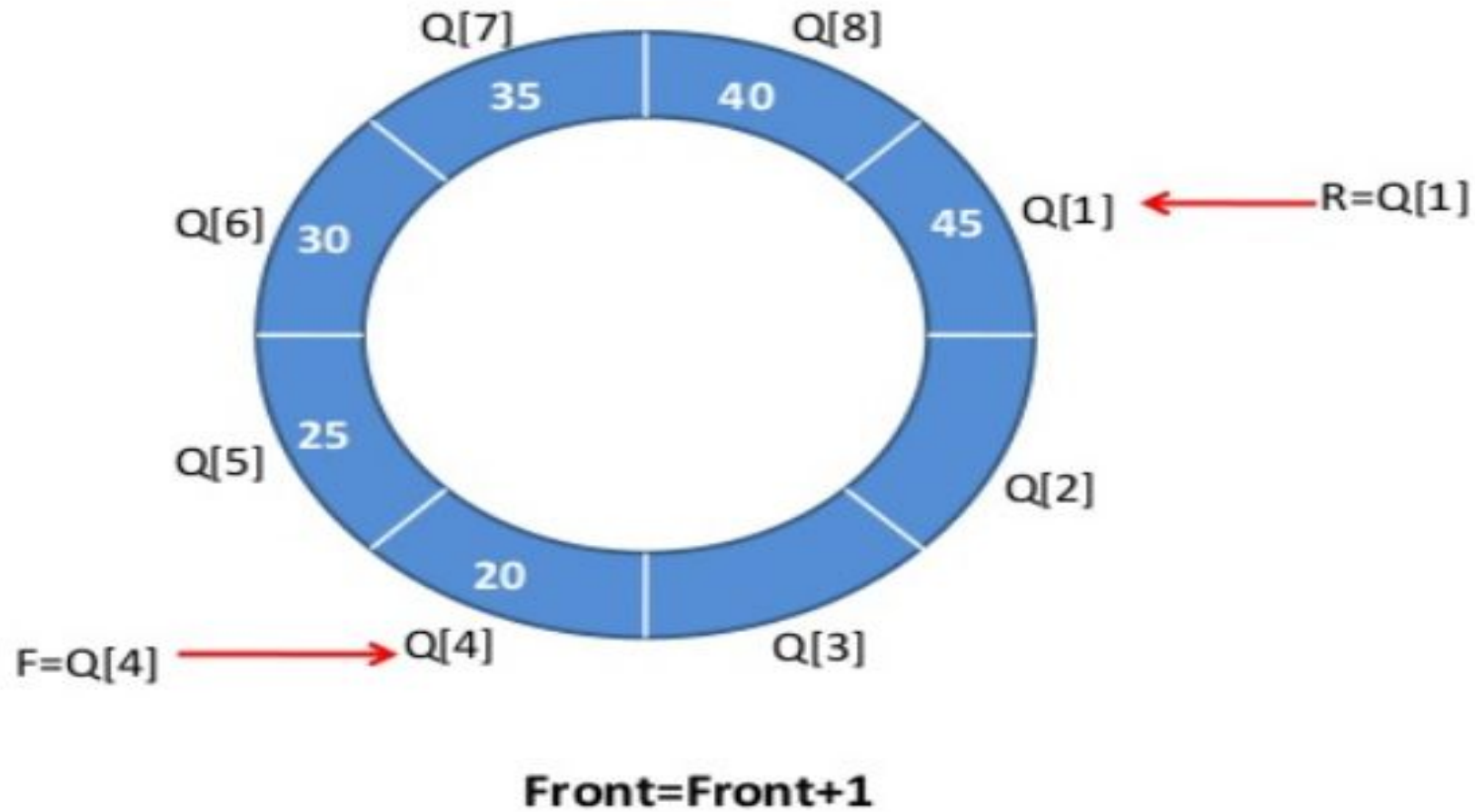
If $F=R=0$ then $F=R=-1$

Deque Operation



If Front=MAX-1 then Front=0

Deque Operation

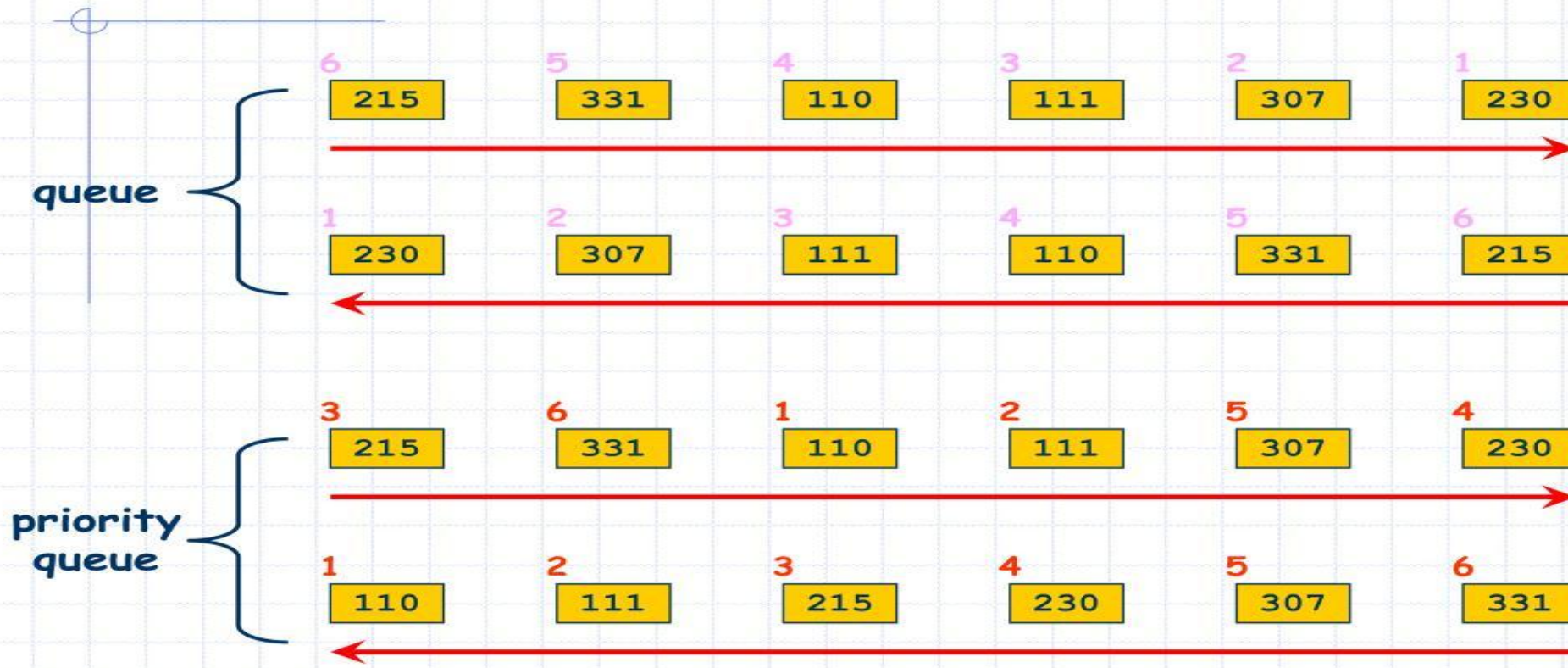


Priority Queue

- A Priority Queue is a queue in which insertion and deletion of items from any position in the queue are done based on some property (such as priority of task)
- Priority Queue contains data items which have some preset priority. While removing an element from a priority queue, the data item with the highest priority is removed first.
- In a priority queue, insertion is performed in the order of arrival and deletion is performed based on the priority.

Priority Queue

Queue vs Priority Queue



Priority Queue – Rules

Rules

1. An element with high priority will be processed before an element with a lower priority.
2. Two elements with the same priority are processed on a First come First Serve basis.

Advantages

1. Easy to Implement
2. Processes with different priority can be efficiently handled.

Deque- Double Ended Queue

- A list in which elements can be inserted or deleted either end.
- It is also known as “Head-Tail Linked List”.
- It has two pointers LEFT and RIGHT, which point to either end of the deque.
- Deque can be implemented using Circular Array or a Circular Doubly Linked List where Deque[N-1] is followed by Deque[0].

Deque – Types

Input Restricted Deque:

Insertion can be done only at one of the ends, while deletion can be done from both ends.

Output Restricted Deque:

Deletion can be done only at one of the ends, while insertions can be done at both ends.

Multiple Queue

SINGLE-LINE QUEUE VS MULTIPLE-LINE QUEUES



Queue Applications

1. Resource sharing among Multiple Consumers-CPU Scheduling, Disk Scheduling
2. When data is transferred asynchronously between two processes – IO Buffers, Pipes, File IO
3. Operating System often maintain a queue of processes that are ready to execute or that are waiting for a particular event to occur
4. Computer Systems often provide a “holding area” for messages between processes/Programs/Systems – Buffer is implemented as queue.
5. Call Center Phone System use queue to hold people in line until a service representative is free.
6. MP3 Players, iPods, CD Players. Newly added songs takes the last position.
7. Round Robin Algorithm on time sharing system is the best example for Circular Queue.

External References

1. <https://nptel.ac.in/courses/106102064/>
2. <https://nptel.ac.in/courses/106103069/>
3. <https://nptel.ac.in/courses/106106127/>
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5. <https://www.geeksforgeeks.org/data-structures/>
6. <https://www.javatpoint.com/data-structure-tutorial>
7. <https://www.coursera.org/learn/data-structures>
8. <https://www.studytonight.com/data-structures/introduction-to-data-structures>
9. <https://www.youtube.com/watch?v=egb6qFw42JM> (Tamil videos)

Thank You